## Mohr's Circle Terminology helpful for Exam 4

**Here are some terms you should be familiar with:**

**"Cross Section":** This is the original area or shape.

**"Reference Points":** Reference points are always the points on the Mohr's Circle which correspond to the various axes (x-y, P-Q, S-T, J-K, etc.) which may exist on the original area.

**"Principal Axes":** These are the principal axes (for Imax and Imin) on the original area.

**"Minor Principal Axis":** This is the $I_{MIN}$ axis. The angle from Ix to $I_{MIN}$ is $q_{p2}$. Angles on MC are + for CCW, and negative for CW.

**"Major Principal Axis":** This is the $I_{MAX}$ axis. The angle from Ix to $I_{MAX}$ is $q_{p1}$. Angles on MC are + for CCW, and negative for CW.

**Axes are always orthogonal (at right angles) to one another:** The problems do not always state it, but any time a set of axes on the "cross section" or on the original area are mentioned (e.g. x-y, P-Q, S-T, J-K, etc.) they are always orthogonal (at right angles to one another), as they should be. The Statics teacher who makes up these problems likes to use odd axes labels.

**Axes may be rotated from the standard x-y orientation:** The arbitrary axes (rotated coordinate system) can generally be rotated at some general angle from the standard x-y (horiz-vertical) orientation. There are generally three axes rotations that are common:

The standard x-y (horizontal-vertical) orientation (Ix, Iy, and Ixy).

The principal orientation (at whatever angle $I_{MIN}$ and $I_{MAX}$ occur on the shape or cross section). Iuv (POI) is zero on this principal coordinate system.

Any other arbitrary orientation (I often call these Iu and Iv, and there is a corresponding Iuv) at any general angle.

**Many of the problems give the Center or the "Average" MOI.** If this is given, always label the center on the picture.

**Some of the problems give a Radius or a Diameter.** If you're given the diameter, halve it to get the radius. This plus the Center defines the Mohr's Circle.

**Always remember that theta values on the "cross section" are drawn as 2*theta on the Mohr's Circle. If an angle is given, read the problem carefully:**

**Is the angle given on the "cross section"?** Then double it when drawing it on the MC.

**Is the angle given for the Mohr's Circle?** Then it can be used "as is" on the MC, but it must be halved when you go back to the "cross section."
If you work on Statics or on studying for Exam 4 over the break, feel free to email me if you have any questions.